

Exam Project

EXAM FORMAT

The exam for *Introduction to Web Applications* consists of two closely connected parts, both of which are mandatory:

1. Individual design and development of a web application
2. Demonstration and oral discussion of the project.

The evaluation of the projects and the demonstration/oral discussion will concern the material submitted before the exam session. The demonstration and brief oral discussion regarding the project must be carried out according to the schedule that will be made available a few days before the exam session, after registration for the exam session itself.

The discussion will focus on the design choices, such as layout, code structure, database structure, etc., as well as the implementation and functional choices adopted.

DESCRIPTION AND REQUIREMENTS

The project consists of a web application that must satisfy a number of technical, stylistic, and functional requirements, detailed below.

The web application must use the technologies illustrated and tested during the course.

Logistical Requirements

- The project must be carried out individually.
- A submission that is partially or entirely shared with another student enrolled in the course is neither allowed nor acceptable.
- The project must be submitted according to the deadlines indicated on the last page of this document.
- Late submissions are neither allowed nor acceptable.

Technical requirements

The web application must comply with the following technical requirements:

- Use of HTML5 and CSS3, relying, if necessary, on external frameworks such as Bootstrap, while customizing their style through ad hoc rules.
- Use of Flask and SQLite, as a relational database, for the back end.
- Use of Flask-Login for authentication management.

In addition:

- The web application must have a clearly defined target device type, chosen by the students: desktop, mobile, or tablet, optionally supporting responsive design.
- Any forms used in the project must be validated both on the front end and on the back end, as explained during the course.
- The submitted project must be fully testable by the instructor and must work on the latest versions of Chrome, 111+, and Firefox, 110+.
- The source code must be well written and include appropriate comments where necessary.
- All the technologies listed above must be integrated cohesively and consistently within a single web application.

Required:

- Deploy the web application on PythonAnywhere: <https://www.pythonanywhere.com/>

Stylistic Requirements

The web application must comply with the following stylistic requirements:

- Use HTML tags semantically, for example, not everything should be a `<div>`.
- Do not use deprecated HTML tags.
- Do not use inline CSS declarations; CSS rules must be kept separate from the semantic structure of the HTML.

In addition, the web application must be sufficiently usable.

Description: Functional requirements

The goal is to create a web application for managing the activities of a cooking school. The school offers different cooking classes, each dedicated to a specific cuisine, recipe, or culinary technique. Each class may be offered several times during the week through scheduled class sessions. Students are free to define the school name, cooking classes, cuisines, chefs, kitchens, and general style of the platform.

The web application must support two types of registered users: **cooking school managers** and **students**.

Cooking school managers can create and manage the cooking classes offered on the platform. Before acting as a manager, the user must register and log in. During registration, the manager must provide their first name, last name, and email address. The email address must be unique and will be used to identify the user on the website.

Students can browse the available cooking classes and enroll in one or more class sessions, as long as their schedules do not overlap. Before enrolling, students must register and log in. During registration, each student must provide their first name, last name, and email address. The email address must be unique and will be used to identify the user on the website.

A student cannot become a cooking school manager, and a cooking school manager cannot enroll in class sessions as a student.

Each cooking class is created and managed by a single cooking school manager and must include the following mandatory information:

- Title of the class.
- Cooking school manager who created and manages the class.
- Cuisine, for example Italian, Japanese, Mexican, Indian, or French.
- Difficulty level: Beginner, Intermediate, or Advanced.
- Duration, expressed in minutes.
- Dietary category: Standard, Vegetarian, Vegan, or Gluten-free.
- Name of the chef who will teach the class.
- List of main ingredients (at least 4).
- Brief description of the class.
- A set of 3 promotional photos.

By clicking on a class session, users must be able to view all the information about the cooking class and the selected session, including the number of places still available.

After logging in, a student can enroll in a class session. Each enrollment is made only for the student themselves, and a student cannot enroll more than once in the same session.

Each student can be enrolled in at most **4 class sessions** during the week and cannot enroll in sessions whose schedules overlap. The system must also prevent enrollments that exceed the maximum number of students allowed for the selected session.

Enrollments can be cancelled only if the cancellation is made at least **12 hours before** the starting time of the class session. After this deadline, the enrollment can no longer be cancelled.

For simplicity, the cooking school operates during a fictional single week, from Monday to Sunday. The application must use a **simulated current day and time** within this week to determine past and upcoming sessions and whether an enrollment can still be cancelled.

Once an enrollment has been completed, it must be visible on the student's profile page. The profile page must show the class title, cuisine, day, starting time, duration, kitchen, and chef. Past and upcoming class sessions must be clearly distinguishable.

Cooking school managers must have a profile page where they can view all the cooking classes they have created and their associated sessions. For each session, they must be able to see the list of enrolled students, the total number of enrolled students, and the number of places still available.

The application must also include a post-class rating functionality. For each class session that has already taken place, students who were enrolled in that session can rate their experience with a score from **1 to 5**. Each student can submit only one rating per class session. For each cooking class that has received at least one rating, the application must display its average rating, considering ratings from all its class sessions.

Note: upon submission, the web application must contain some sample data allowing the instructors to test its main functionalities. In particular, it must include at least two cooking school managers, four students, several cooking classes with different cuisines, difficulty levels, durations, and dietary categories, and several class sessions across different days and kitchens. It must also include existing enrollments, at least one fully booked session, at least one session with no enrollments that can still be modified, both past and upcoming sessions, and some ratings. The usernames and passwords of the sample accounts, together with the simulated current day and time, must be provided to the instructors for verification.

🍳 Students are encouraged to fully express their creativity in choosing the cooking school name, cuisines, class topics, recipes, chefs, kitchens, visual identity, images, and the overall experience offered to users. The goal is to create an original and personalized project.

OPTIONAL EXTRA REQUIREMENT FOR PROVA FINALE

Students who need to complete the **Prova finale** may extend the project by implementing a **waiting list functionality** for fully booked class sessions.

When a class session reaches its maximum capacity, a student who would otherwise be allowed to enroll can join its waiting list. The waiting list must follow a **first come, first served** policy.

Students must be able to view the class sessions for which they are on a waiting list, their current position in each list, and leave a waiting list. A student cannot simultaneously be enrolled in and be on the waiting list for the same session. Joining a waiting list does not count toward the maximum of 4 class session enrollments.

When an enrolled student cancels and a waiting list exists, the first student on the waiting list must be **automatically enrolled**, and the remaining students must move forward accordingly.

Cooking school managers must be able to view the waiting list for each of their class sessions. They must also be able to view basic statistics about the activities they manage, such as:

- Total number of cooking classes and class sessions.
- Total number of enrollments.
- Total number of students currently on waiting lists.
- Most popular cuisine, based on enrollments.
- Cooking class with the highest average rating.

This additional functionality is optional and is required only for students completing the **Prova finale**. It should be integrated into the same web application in a coherent way, without replacing or simplifying the mandatory requirements described above.

PROJECT SUBMISSION INSTRUCTIONS

The project must be submitted through the “**Consegna Elaborati**” tool available on the course page on the Teaching Portal, by **23:59 on 13 September 2026**.

The submission must contain a single archive file, **.zip**, including:

- Source code of the developed web application, including:
 - any dependencies, in a file such as **requirements.txt**;
 - images;
 - etc.;
 - the SQLite file containing the database.
- A text document, **.md** or **.txt**, containing:
 - the user credentials;
 - any instructions needed to test the web application;
 - the url where the deployed web application is available.